

Problem 3.10

Given $A(0) = \$1,500$ and $A(15) = \$2,700$. What is i assuming simple interest?

Solution

Under simple interest,

$$A(t) = P(1 + it).$$

Since

$$A(0) = 1500 = x(1) = 1500,$$

we have

$$P = 1500.$$

Now use $A(15) = 2700$:

$$A(15) = 1500(1 + 15i).$$

Thus,

$$\frac{2700}{1500} = 1 + 15i.$$

So,

$$i = \frac{\frac{2700}{1500} - 1}{15}.$$

Therefore,

$$i = 0.0533 \approx 5.33\%.$$